Phase-only axial intensity shaping in a scalar band-limited propagation model: a simulation-only screening study

AutoPhotonics Project

Abstract

We study whether a passive phase-only modulation of an equal-power Gaussian beam can raise the minimum on-axis intensity over a pre-fixed axial window of 49.0–71.0 mm relative to an ideal focusing lens. The model is a scalar, single-wavelength, band-limited angular-spectrum propagation on a 128×128 grid with a 2.40 mm square window, $\lambda = 633$ nm, $w_0 = 300\,\mu$m, $f = 60$ mm and $P = 1$ W. We compare a Gaussian-plus-lens reference, an optimized one-parameter defocus control, an optimized quadratic-plus-quartic control, and a four-parameter mixed polynomial-plus-ripple candidate found by a seeded hybrid search. At the 15 sampled planes, the best candidate reaches $J = 0.51366$ versus 0.34640 for optimized defocus and 0.46550 for the quadratic-plus-quartic control, corresponding to +48.29% and +10.35% in the fixed objective. Its relative peak is 0.61028, so the gain is a worst-case/peak tradeoff. Grid refinement changes J by -4.04×10^{-6} relative, and idealized 8-bit phase quantization changes it by -9.38×10^{-4} relative. The study is simulation-only: it does not establish prior-art novelty, independent-solver validation, realistic spatial-light-modulator performance, or experimental verification.

1 Introduction

Phase-only masks are a standard route to modify focusing without changing input power. Wavefront coding and extended-depth-of-focus systems have used annular, cubic and related phase profiles for imaging and beam-shaping tasks [2, 5, 3]. The present study asks a narrower question: within a scalar free-space model and a pre-fixed axial interval, can a passive residual phase on an equal-power Gaussian input increase the minimum on-axis intensity compared with an ideal focusing lens?

The question is deliberately separated from a novelty claim. We report a numerical screening result, compare it with optimized controls, and state the checks and missing validations. The work does not establish that the candidate phase family is new, that the gain survives other axial windows, or that it transfers to a realistic spatial light modulator (SLM). The related citations are metadata-verified context, not full-text prior-art evidence.

2 Theoretical model

2.1 Field, coordinates and normalization

The field is a scalar amplitude $U(x, y, z)$ on a finite square computational window of side L. The transverse coordinates are x and y, the propagation coordinate is z, and intensity is $I = |U|^2$. The grid has n points per side, $\Delta x = L/n$, and grid indices $j, k = 1, \ldots, n$. The input power is P. The Gaussian waist is w_0, the vacuum wavelength is λ, and $k_0 = 2\pi/\lambda$. Lengths are in metres, spatial frequencies in m^{-1}, phases in radians, and P in watts. The objective quantities J, J_{peak}, V and R are dimensionless.

The input field combines a Gaussian amplitude, an ideal thin-lens phase and a residual phase ϕ_{jk}:

$$\begin{aligned} A_{jk} &= \sqrt{\frac{P}{\Delta x^2 \sum_{a,b} \exp(-2r_{ab}^2/w_0^2)}}\, \exp(-r_{jk}^2/w_0^2), \\ U_0(x_j, y_k) &= A_{jk} \exp\left[i \left(-\frac{k_0 r_{jk}^2}{2f} + \phi_{jk} \right) \right]. \end{aligned} \tag{1}$$

Here $r_{jk}^2 = x_j^2 + y_k^2$, f is the ideal lens focal length, and the prefactor A_{jk} enforces the discrete power normalization $\Delta x^2 \sum_{j,k} |U_0(x_j, y_k)|^2 = P$. The lens phase is ideal and is not quantized. The residual phase is the only designed degree of freedom.

The residual phase is written in normalized transverse coordinates

$$u = x/w_0, \quad v = y/w_0, \quad \rho^2 = u^2 + v^2, \qquad \phi = q_2\rho^2 + q_3(u^3 + v^3) + q_4\rho^4 + \eta \sin(2\rho^2), \tag{2}$$

with coefficients q_2, q_3, q_4 and ripple coefficient η in radians. The term $q_2\rho^2$ is defocus-like, $q_4\rho^4$ is a quartic radial term, $q_3(u^3 + v^3)$ breaks radial symmetry, and $\eta \sin(2\rho^2)$ is a deterministic radial ripple. This polynomial family is a low-order residual-phase parameterization; it is not a normalized Airy or Pearcey beam. The phase in Eq. (2) is not separately normalized to zero mean; power normalization is carried by Eq. (1).

2.2 Propagation

Propagation uses the band-limited angular-spectrum transfer function

$$\begin{aligned} H_z(\nu_x, \nu_y) &= B_z \exp\left[i 2\pi z \sqrt{\lambda^{-2} - \nu_x^2 - \nu_y^2} \right], \\ B_z &= \mathbf{1}_{\nu_x^2 + \nu_y^2 \le \min(\nu_c^2, \lambda^{-2})}, \\ \nu_c &= \frac{\lambda^{-1}}{\sqrt{1 + (2z/L)^2}}. \end{aligned} \tag{3}$$

The square root is evaluated on the propagating passband, and H_z is zero outside it. B_z is one on the passband and zero outside. The radial bandwidth truncation in Eq. (3) is conservative; it is not the complete Matsushima passband construction [4]. The discrete propagation and on-axis intensity are

$$U_z = \mathcal{F}^{-1}[\mathcal{F}[U_0] H_z], \qquad I(z) = |U_z(0,0)|^2, \tag{4}$$

where $\mathcal{F}$ is the discrete FFT and $\mathcal{F}^{-1}$ its matching inverse. All reported on-axis values are evaluated at the central grid point $(0, 0)$ without transverse interpolation.

2.3 Objective, spectral retention and quantization

The axial window is the finite set $\mathcal{Z} = \{z_m\}$ of sampled planes, not a continuous interval. The objective normalizes the candidate on-axis intensity by the maximum on-axis intensity of the Gaussian-plus-lens reference:

$$I_{\mathrm{ref}} = \max_{z \in \mathcal{Z}} I_{\mathrm{G}}(z), \qquad J = \frac{\min_{z \in \mathcal{Z}} I(z)}{I_{\mathrm{ref}}}, \quad J_{\mathrm{peak}} = \frac{\max_{z \in \mathcal{Z}} I(z)}{I_{\mathrm{ref}}}, \quad V = \frac{J}{J_{\mathrm{peak}}}. \tag{5}$$

J is the worst-case sampled on-axis intensity relative to the Gaussian peak, J_{peak} is the corresponding relative peak, and V is a uniformity ratio bounded by one. The search maximizes J; it does not maximize total transmitted power, spot size, resolution, trapping force or any application-level quantity.

Spectral retention is measured as the minimum passband energy fraction over the sampled planes:

$$R = \min_{z \in \mathcal{Z}} \frac{\sum_{a,b} |\widehat{U}_{0,ab}|^2 B_{z,ab}}{\sum_{a,b} |\widehat{U}_{0,ab}|^2}, \tag{6}$$

where $\widehat{U}_{0,ab} = \mathcal{F}[U_0]_{ab}$. Because the radial mask is conservative, R is close to unity for all designs; the lost fraction $1 - R$ is reported in scientific notation rather than treating R as exactly one.

Quantization is applied only to the wrapped residual phase:

$$\phi_8 = \frac{2\pi}{255} \operatorname{round}\left[\frac{255}{2\pi}(\phi \bmod 2\pi)\right]. \tag{7}$$

This is an idealized 8-bit wrapped-phase model. The lens phase remains ideal, and no SLM calibration, fill factor, pixel pitch, aperture apodization, noise, aberration or photodamage constraint is included.

3 Numerical setup

3.1 Shared conditions

The run uses $n = 128$, $L = 0.0024\,\mathrm{m}$, $\lambda = 6.33 \times 10^{-7}\,\mathrm{m}$, $w_0 = 3.0 \times 10^{-4}\,\mathrm{m}$, $f = 0.06\,\mathrm{m}$ and $P = 1.0\,\mathrm{W}$. Thus $\Delta x = L/n = 18.75\,\mu\mathrm{m}$. The axial window is

$$z_m = 0.049 + m\,\frac{0.071 - 0.049}{14}\ \mathrm{m}, \qquad m = 0, \ldots, 14, \tag{8}$$

with step $0.00157142857142857\,\mathrm{m}$; the last plane is $z = 0.071\,\mathrm{m}$. The 15 sampled planes and the objective in Eq. (5) were fixed before screening. No interpolation in z or in the transverse plane is used. The Gaussian reference, the optimized controls and the candidate are propagated with the same grid, window, normalization and transfer function.

3.2 Controls and search protocol

The Gaussian reference uses $q_2 = q_3 = q_4 = \eta = 0$ with the fixed ideal lens. The optimized defocus control varies q_2 only. The optimized quadratic-plus-quartic control varies q_2 and q_4. The mixed candidate family varies q_2, q_3, q_4 and η. The common bounds are $q_2 \in [-1.2, 1.2]$, $q_3 \in [-0.4, 0.4]$, $q_4 \in [-0.7, 0.7]$ and $\eta \in [-1.0, 1.0]$. All arms use seed 7 and the same optimizer: shared nested-family initialization followed by bounded differential mutation, jitter and seeded restarts.

Each optimized control arm used 1024 proposal attempts. The hybrid arm used 1019 numerical proposals and 5 language-model proposals. The proposal-attempt budgets are matched, but the hybrid additionally sees the optimized control outcomes. This is informed augmentation, not an equal-information comparison. Only five of the 1024 hybrid proposals were language-model proposals; the run does not establish an LLM-specific benefit. Parameter ablation, multi-seed sampling, holdout optimization and alternate-window sensitivity were not available in the run record.

4 Results and discussion

4.1 Optimized controls and candidate

Table 1 reports the fixed-window metric for the Gaussian reference and the three optimized designs. Table 2 lists the phase coefficients. The strongest optimized control is the quadratic-plus-quartic design, not the one-parameter defocus control.

Table 1: Sampled-window metrics. J is the minimum on-axis intensity normalized to the Gaussian peak, J_{peak} is the corresponding maximum, $V = J/J_{\text{peak}}$, and the last column is the lost passband fraction $1 - R$ from Eq. (6).

Design	J	J_{peak}	V	Lost fraction
Gaussian + ideal lens	0.302247	1.000000	0.302247	1.744×10^{-10}
Optimized defocus	0.346402	0.968147	0.357799	7.685×10^{-11}
Optimized quadratic + quartic	0.465503	0.695010	0.669779	2.828×10^{-11}
Best mixed candidate	0.513664	0.610283	0.841682	6.573×10^{-11}

Table 2: Phase coefficients in radians for the normalized coordinates of Eq. (2). The Gaussian reference has all coefficients zero.

Design	q_2	q_3	q_4	η
Gaussian + ideal lens	0	0	0	0
Optimized defocus	0.136916	0	0	0
Optimized quadratic + quartic	−1.002303	0	0.656616	0
Best mixed candidate	−0.442000	0.014000	0.378000	−0.606900

The candidate raises J from 0.346402 for optimized defocus to 0.513664, a relative increase of +48.29%. Relative to the stronger quadratic-plus-quartic control, it raises J from 0.465503 to 0.513664, a relative increase of +10.35%. These percentages refer only to the fixed objective J. They are not improvements in total transmitted power, spot size, resolution, trapping force or overall research impact.

The candidate also lowers the relative peak: $J_{\text{peak}} = 0.610283$ versus 0.968147 for defocus and 0.695010 for the quadratic-plus-quartic control. Its uniformity ratio is higher ($V = 0.841682$ versus 0.669779 for the quartic control), which is consistent with a flatter sampled axial profile at the cost of peak intensity. The Gaussian reference has $V = 0.302247$ because its on-axis intensity varies strongly across the fixed window. The tradeoff is therefore explicit: the candidate improves the worst-case sampled value while reducing the maximum.

Figure 1 shows the normalized on-axis intensity over the sampled window. The curves are discrete samples at the 15 planes of Eq. (8); the plotted values are those used in Eq. (5). No smoothing or axial interpolation is applied. The candidate's higher floor and lower peak are visible as the two sides of the same objective tradeoff.

Figure 2 shows the residual phase of the candidate and a common-scale intensity comparison at focal planes. The phase panel displays the mixed polynomial-plus-ripple residual phase from Eq. (2); the intensity panel uses a common scale so that the relative peak differences in Table 1 can be compared directly. The figure is a diagnostic view, not an independent validation of the propagation model.

4.2 Numerical checks and interpretation

Table 3 lists the available numerical checks. A higher-grid run with $n = 192$ gives $J = 0.513662$, a relative change of -4.04×10^{-6} from the 128×128 value. Idealized 8-bit quantization of the wrapped residual phase gives $J = 0.513183$, a relative change of -9.38×10^{-4} (-0.0938%). The power fraction at the window edges on the final propagation plane is 4.14×10^{-9}, and the candidate has 0.977565 of its power within a $110\,\mu$m radius at the nominal focal plane.

The spectral retention in Eq. (6) is close to unity for every design because the conservative radial passband includes almost all of the input spectrum. For the candidate, the lost fraction is 6.57×10^{-11}. The earlier rounded display of $R = 1.000000$ should therefore be read as near-unity retention, not exact unity. This check confirms that the conservative mask removes a

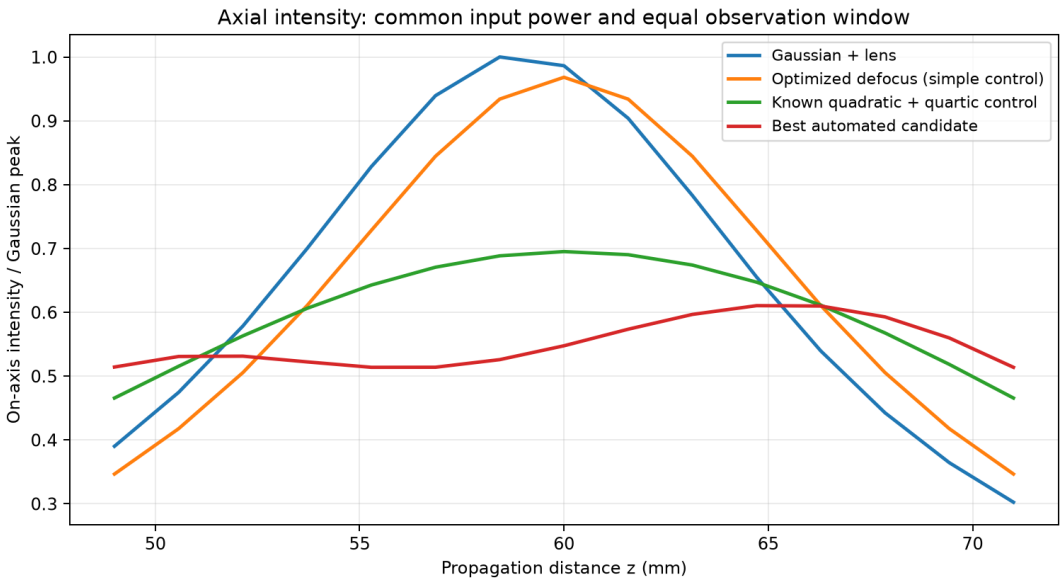


Figure 1: Normalized on-axis intensity over the fixed sampled axial window for the Gaussian-plus-lens reference, optimized defocus, optimized quadratic-plus-quartic control and best mixed candidate. The plotted values are sampled at the 15 planes of Eq. (8) and normalized by the Gaussian peak in Eq. (5). The run is a single seed; no error bars or interpolation are shown.

negligible fraction of the sampled input spectrum; it does not validate the propagation against an independent solver or the complete passband construction.

The mixed mask changes the phase distribution and can redistribute energy among axial positions. The observed increase in the minimum sampled on-axis intensity is consistent with such redistribution, but the present comparisons do not isolate which phase term causes it. The candidate has more parameters than the defocus control and more parameters than the quadratic-plus-quartic control; without parameter ablation or matched-capacity controls, the +10.35% gain over the quartic control may reflect added flexibility rather than a distinct physical phase-mask effect. This is an interpretation, not a causal claim established by ablation.

The focused-power fraction of 0.977565 within 110 μm is measured at the nominal focal plane. It is not the fixed-window worst-case objective and does not guarantee performance across the window. Likewise, the edge-power fraction of 4.14×10^{-9} shows negligible power at the final window boundary in this run; it is not a general aperture or device constraint.

4.3 Limitations

The study is single-seed and sampled-z. No multi-seed variability, uncertainty estimate, holdout optimization, alternate axial window or parameter ablation was run. The fixed window and metric were chosen before screening, which reduces but does not eliminate overfitting risk; the absence of alternate-window tests means the stability of the gain is unknown. The comparison is not matched in model capacity: the candidate has four parameters, the quartic control has two and the defocus control has one. The hybrid arm sees the optimized control outcomes, so matched proposal-attempt budgets do not establish matched information. The five language-model proposals are too few to support an LLM-specific claim.

The model is highly idealized. It is scalar and linear, with a single wavelength, an ideal thin lens, a square window and a conservative radial bandlimit. It does not include polarization, nonlinearity, fractional diffraction, anisotropic media, SLM phase calibration, fill factor, pixel pitch, aperture apodization, noise, aberrations or photodamage constraints. The idealized 8-bit phase quantization is not a hardware model. Scalar intensity alone does not determine optical

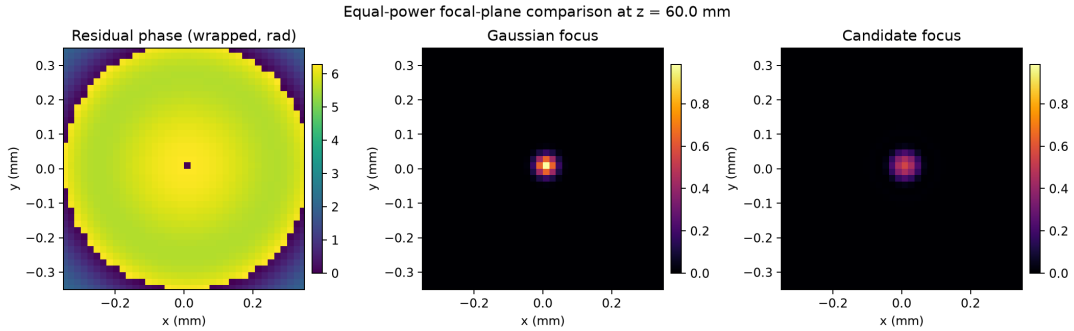


Figure 2: Residual phase of the best mixed candidate and common-scale intensity comparison at focal planes. The phase follows Eq. (2) with the coefficients in Table 2; the intensity comparison uses the same normalization as Table 1. The ideal lens phase is not included in the residual-phase panel.

Table 3: Numerical checks for the best candidate. The grid and quantization entries are relative changes in J; the edge entry is a power fraction on the final plane; the focused-power entry is a fraction within a 110 μm radius at the nominal focal plane. These are numerical screening checks, not independent validation.

Check	Observed
Higher-grid size	$n = 192$
Higher-grid J	0.513662
Relative change with grid refinement	-4.04×10^{-6}
Quantized J	0.513183
Relative change after 8-bit phase quantization	-9.38×10^{-4}
Edge power fraction at last z	4.14×10^{-9}
Focused power fraction within 110 μm	0.977565
Numerical performance gates	pass (screening only)

trapping forces or material response. No axicon, cubic phase plate or published extended-depth-of-focus mask was optimized as an additional baseline under matched budgets; the cited phase-mask literature [2, 5, 3] is context, not a complete prior-art survey.

Prior-art novelty is not established. The candidate components—defocus, quartic aberration, a cubic-like transverse term and a radial ripple—are familiar in wavefront coding, focusing and extended-depth-of-focus work. The citations used here were metadata-verified, but no full-text prior-art comparison or human-verified assessment of the closest papers was performed. The numerical performance gates pass only for the stated screening checks; they do not establish publication readiness or experimental relevance.

5 Conclusions

Within the stated scalar model and the pre-fixed sampled window of 49.0–71.0 mm, the best mixed phase-only candidate reaches a minimum normalized on-axis intensity of $J = 0.513664$, compared with 0.346402 for optimized defocus and 0.465503 for an optimized quadratic-plus-quartic control. The corresponding relative gains are +48.29% and +10.35% in the fixed objective. The candidate does so with a lower relative peak (0.610283 versus 0.695010 for the quartic control) and a higher uniformity ratio (0.841682 versus 0.669779). Grid refinement and idealized 8-bit phase quantization change J by small relative amounts, and the conservative passband retains nearly all input spectral energy.

These results support a screening statement, not a discovery claim. They do not establish prior-art novelty, an independent-solver validation, a realistic SLM implementation, alternate-window robustness, matched-capacity attribution or experimental verification. The following work is required before any stronger claim: full-text prior-art comparison with human verification; an independent wave-optics solver or analytical limiting-case check; optimized axicon, cubic and published extended-depth-of-focus baselines under matched budgets; parameter ablation and perturbation tests; multi-seed, holdout and alternate-window sensitivity; a realistic SLM model with calibration, fill factor, pixel pitch, aperture, noise and aberrations; and, if an application is intended, explicit calculation of optical forces and material response.

Disclosures

This manuscript was generated by AutoPhotonics Scientist, which adapts Sakana AI Scientist-v2 [1]. The work is reported under the combined status **SIMULATION_ONLY**, **NOVELTY_NOT_VERIFIED** and **NOT_PUBLICATION_READY**. It is a numerical screening study: no experiment, independent propagation solver, realistic SLM characterization, human-verified full-text prior-art comparison or external peer review was performed. The automatic review is an AI assessment; human review and external submission remain separate.

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